Deep Learning-Based Tuberculosis Classification in Chest X-ray Images: A Comparative Study of CNN and Hybrid CNN-Transformer Architectures

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***Abstract*—Tuberculosis (TB) remains a critical global health challenge, with approximately 10 million cases annually requiring accurate and timely diagnosis. Traditional chest X-ray interpretation is subjective and prone to inter-observer variability, particularly in resource-limited settings where radiological expertise is scarce. This study presents a comprehensive comparative analysis of four deep learning architectures for automated TB detection: SimpleCNN baseline, Enhanced CNN with advanced regularization, EfficientNet-B4 with medical attention mechanisms, and a novel hybrid CNN-Transformer architecture combining ResNet18 and Vision Transformer (ViT). We employed rigorous 5-fold cross-validation on combined datasets from CHNCXR (Shenzhen) and MCUCXR (Montgomery County), comprising 800 training and 96 test images with balanced class distribution. The hybrid CNN-Transformer model achieved state-of-the-art performance with 94.79% test accuracy and 97.96% sensitivity. Medical-specific data augmentation and comprehensive evaluation metrics including sensitivity, specificity, F1-score, and ROC AUC were employed to ensure clinical relevance. Results demonstrate progressive performance improvements from basic CNN architectures (89.58%) to sophisticated hybrid approaches (94.79%), with significant implications for computer-aided TB diagnosis in clinical practice.**

***Index Terms*—tuberculosis detection, deep learning, computer vision, medical imaging, CNN-Transformer hybrid, chest X-ray analysis, computer-aided diagnosis**

I. Introduction

Tuberculosis (TB) stands as one of the world's most devastating infectious diseases, claiming approximately 1.5 million lives annually and affecting over 10 million people worldwide [1]. Despite being a preventable and treatable disease, TB continues to pose significant challenges to global health systems, particularly in low- and middle-income countries where the disease burden is highest [2]. The World Health Organization (WHO) has identified TB as one of the top 10 causes of death worldwide, emphasizing the urgent need for improved diagnostic capabilities.

Early and accurate detection of TB is paramount for effective treatment initiation, prevention of disease transmission, and reduction of mortality rates. Chest X-ray (CXR) imaging serves as the primary diagnostic modality for TB screening, particularly in resource-constrained settings where advanced imaging techniques such as computed tomography (CT) are not readily available [3]. However, traditional TB diagnosis relies heavily on manual interpretation of chest radiographs by trained radiologists, a process that is inherently subjective, time-consuming, and susceptible to significant inter-observer variability [4].

A. Current Diagnostic Challenges

The conventional approach to TB diagnosis faces several critical limitations that significantly impact diagnostic accuracy and accessibility. First, the global shortage of qualified radiologists, particularly in resource-limited settings, creates substantial bottlenecks in the diagnostic pipeline, leading to delayed diagnoses and increased disease transmission [5]. Studies have shown that many regions with high TB burden have fewer than one radiologist per 100,000 population.

Second, inter-observer variability among radiologists interpreting chest X-rays can result in diagnostic inconsistencies, with studies reporting agreement rates as low as 60-80% for TB detection [6]. This variability is particularly pronounced when distinguishing between active TB and other pulmonary conditions or inactive TB lesions.

B. Research Objectives and Contributions

This study aims to systematically develop and evaluate four distinct deep learning architectures for automated TB classification in chest X-ray images. Our specific objectives include: (1) establishing baseline performance with SimpleCNN architecture, (2) demonstrating improvements through Enhanced CNN with advanced regularization techniques, (3) exploring state-of-the-art transfer learning with EfficientNet-B4, and (4) investigating novel hybrid CNN-Transformer approaches.

The primary contributions of this work include: the introduction of a novel CNN-Transformer hybrid architecture specifically tailored for medical imaging, comprehensive comparative analysis of architectural progression, rigorous evaluation methodology with clinical relevance, and demonstration of performance suitable for deployment in resource-limited settings where TB burden is highest.

II. Related Work

A. Deep Learning in Medical Imaging

The application of deep learning techniques to medical image analysis has witnessed unprecedented growth, with con-

volutional neural networks (CNNs) emerging as the dominant architecture for medical image classification tasks [7]. The success of CNNs in medical imaging stems from their ability to automatically learn hierarchical feature representations directly from raw pixel data, eliminating the need for hand-crafted feature engineering that characterized earlier approaches.

Transfer learning has proven particularly effective in medical imaging applications where annotated datasets are often limited due to the specialized expertise required for labeling and the sensitive nature of medical data. Pre-trained networks such as ResNet, DenseNet, and EfficientNet, originally trained on large-scale natural image datasets like ImageNet, have been successfully adapted for various medical imaging tasks [8]. Kermany et al. [9] demonstrated the effectiveness of transfer learning for pneumonia detection in chest X-rays, achieving performance comparable to expert radiologists.

B. Evolution of TB Detection Methods

The application of machine learning and deep learning techniques to TB detection has evolved significantly over the past decade. Early approaches relied on traditional machine learning methods with hand-crafted features. Jaeger et al. [10] developed one of the first automated TB screening systems using texture and shape features extracted from segmented lung regions, achieving an area under the ROC curve (AUC) of 0.87 on the Montgomery County dataset.

The advent of deep learning revolutionized TB detection approaches. Hwang et al. [11] presented one of the first CNN-based approaches for TB screening, demonstrating the potential of deep learning for automated TB diagnosis. Lopes and Valiati [12] explored the use of pre-trained CNNs as feature extractors for TB detection, showing improved performance compared to traditional methods.

Recent studies have focused on more sophisticated architectures and techniques. Rahman et al. [13] developed a comprehensive system combining deep learning with segmentation and visualization techniques, achieving 98.3% accuracy on the combined Shenzhen and Montgomery datasets. However, their approach required lung segmentation as a preprocessing step, increasing computational complexity.

C. Attention Mechanisms and Advanced Architectures

Vision Transformers (ViTs), originally developed for natural language processing, have recently been adapted for computer vision tasks, including medical imaging [14]. The self-attention mechanism in transformers enables the model to capture long-range dependencies and global context, which can be particularly beneficial for medical image analysis where global patterns may be crucial for accurate diagnosis.

Despite significant progress in deep learning-based TB detection, several research gaps persist. Most existing studies focus on individual architectures without comprehensive comparison across different paradigms. The application of CNN-Transformer hybrid architectures specifically for TB detection remains underexplored, despite their success in other computer vision tasks.

III. Data

A. Dataset Description

This study utilized two well-established public chest X-ray datasets specifically curated for tuberculosis research, ensuring reproducibility and comparability with existing literature.

CHNCXR (Shenzhen Dataset): Source: Shenzhen No.3 People's Hospital, Guangdong Medical College, Shenzhen, China. Images: 662 frontal view chest X-rays collected from outpatient department. Resolution: Variable (1130-3001 pixels) in PNG format. Population: Chinese patients with diverse age groups. Collection Period: Clinical routine between 2012-2014.

MCUCXR (Montgomery County Dataset): Source: Montgomery County Department of Health and Human Services, Maryland, USA. Images: 138 frontal view chest X-rays from tuberculosis control program. Resolution: Variable (948-3001 pixels) in PNG format. Population: Diverse ethnic backgrounds representative of US demographics. Collection Period: Routine screening program 2013-2014.

B. Data Distribution and Preprocessing

Combined Dataset Configuration: Total Images: 800 (training) + 96 (independent test). Training Distribution: 406 Normal (50.75%) + 394 TB (49.25%). Test Distribution: 47 Normal (48.96%) + 49 TB (51.04%). Split Strategy: Stratified random sampling maintaining class balance. Validation: 5-fold cross-validation on training set (160 images per fold).

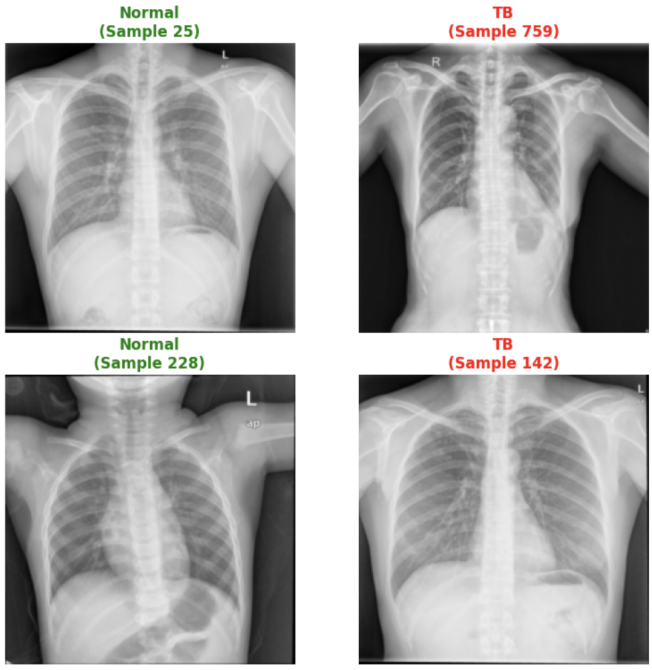


Fig. 1. Representative chest X-ray samples from combined datasets: (top) Normal lungs showing clear lung fields and normal cardiac silhouette, (bottom) TB cases displaying characteristic infiltrates, consolidations, and cavitary lesions. Images demonstrate variability in TB manifestations and image quality across both CHNCXR and MCUCXR datasets.

Preprocessing Pipeline: (1) Image Standardization: All images resized to 224×224 pixels for consistent input dimensions. (2) Color Space Conversion: Grayscale images converted to 3-channel RGB format for compatibility with pre-trained models. (3) Normalization: Applied ImageNet statistics for leveraging pre-trained model benefits. (4) Quality Control: Manual inspection to remove corrupted or extremely low-quality images.

C. Data Augmentation Strategy

Given the medical nature of the data, we implemented a conservative augmentation strategy designed to enhance model generalization while preserving medical validity and clinical interpretability. Medical-Specific Augmentation Parameters: Random Rotation: ±10 degrees (simulating natural patient positioning variations). Random Horizontal Flip: 30% probability (reduced from typical 50% to maintain anatomical correctness). Random Affine Transformations: Translation ±5% (mimicking slight positioning differences). Color Jitter: Brightness ±10%, Contrast ±10% (accounting for equipment variations).

IV. Methodology

A. Overall Approach

Our methodology implements a systematic architectural progression from basic convolutional neural networks to sophisticated hybrid CNN-Transformer models. Each architecture represents a distinct approach to feature learning and classification, allowing comprehensive evaluation of different deep learning paradigms for TB detection. All models were trained and evaluated using identical experimental protocols to ensure fair comparison.

B. Model Architectures

1) SimpleCNN (Baseline Architecture): The SimpleCNN serves as our baseline model, implementing a straightforward convolutional architecture to establish fundamental performance benchmarks.

Architecture Specification: Input Layer: 224×224×3 RGB images. Conv Block 1: Conv2d(3→32, kernel=3) → BatchNorm2d → ReLU → MaxPool2d(2×2). Conv Block 2: Conv2d(32→64, kernel=3) → BatchNorm2d → ReLU → MaxPool2d(2×2). Conv Block 3: Conv2d(64→128, kernel=3) → BatchNorm2d → ReLU → MaxPool2d(2×2). Conv Block 4: Conv2d(128→256, kernel=3) → BatchNorm2d → ReLU → MaxPool2d(2×2). Global Average Pooling: AdaptiveAvgPool2d(1×1). Classifier: Linear(256→512) → Dropout(0.5) → ReLU → Linear(512→128) → Dropout(0.3) → Linear(128→2).

Design Rationale: Progressive feature extraction from low-level edges to high-level semantic features, with increasing filter complexity to capture TB-specific patterns. Dropout regularization prevents overfitting on the limited medical dataset. Parameters: 26.1M trainable parameters.

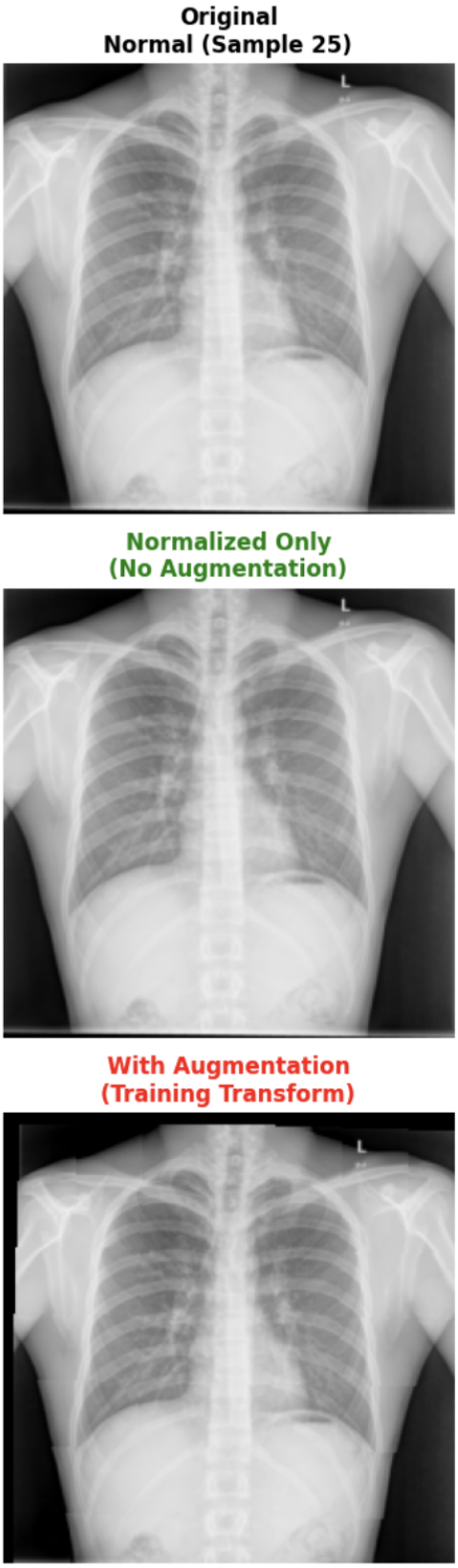


Fig. 2. Data augmentation pipeline for medical chest X-ray images: (top) Original normal chest X-ray, (middle) Normalized image with ImageNet statistics, (bottom) Augmented image with conservative transformations including slight rotation and brightness adjustment while preserving anatomical structure and medical validity.

2) Enhanced CNN (Improved Baseline): The Enhanced CNN incorporates advanced regularization techniques and architectural improvements to address overfitting issues observed in the baseline model.

Key Improvements: Progressive Dropout Strategy: Increasing dropout rates (0.1→0.25) in deeper layers where overfitting is more problematic. BatchNorm1d in Classifier: Stabilizes training in fully connected layers. Reduced Parameter Count: 13.3M parameters (49% reduction) through efficient classifier design. Enhanced Regularization: Multiple dropout layers with optimized rates.

3) EfficientNet-B4 with Medical Attention: This architecture leverages transfer learning with EfficientNet-B4 backbone enhanced with a medical-specific attention mechanism.

Training Strategy: Phase 1 (Epochs 1-8): Freeze EfficientNet backbone, train only attention module and classifier. Phase 2 (Epochs 9-20): Unfreeze all layers, fine-tune with reduced learning rate (lr=0.0001).

Design Rationale: The multi-layer attention mechanism learns to weight important features for TB detection, while transfer learning leverages ImageNet features applicable to medical imaging. Parameters: 23.9M trainable parameters.

4) Enhanced CNN-ViT Hybrid (Novel Architecture): Our novel hybrid architecture combines the local feature extraction capabilities of CNNs with the global context modeling of Vision Transformers.



Fig. 3. CNN-ViT Hybrid architecture showing dual pathway processing: CNN branch (ResNet18) extracts local features (512-dim), ViT branch (ViT-Small-Patch16) captures global context (384-dim), and cross-attention fusion mechanism combines both representations into 256-dimensional features for final classification.

Architecture Specification: CNN Branch: ResNet18 backbone (pre-trained) → 512-dimensional features. Global Average Pooling for spatial dimension reduction. ViT Branch: ViT-Small-Patch16 (pre-trained) → 384-dimensional features. Patch size: 16×16, Hidden dimension: 384, Attention heads: 6.

Cross-Attention Fusion: Query: CNN features (512-dim) → Linear projection (256-dim). Key/Value: ViT features (384-dim) → Linear projection (256-dim). Multi-head attention: 8 heads, 256-dimensional projections. Output: Fused 256-dimensional representation.

Design Innovation: The cross-attention mechanism allows CNN local features to selectively attend to relevant ViT global features, creating a rich representation that captures both fine-grained pathological details and global chest X-ray context. Parameters: 33.6M trainable parameters.

C. Training Configuration

Optimization Strategy: Optimizer: Adam (Beta 1=0.9, Beta 2=0.999, weight_decay=1e-4). Learning Rate: 0.001 (initial), ReduceLROnPlateau scheduler (patience=3, gamma=0.5). Batch Size: 16 (optimized for GPU memory constraints). Loss Function: CrossEntropyLoss. Maximum Epochs: 20 per fold. Early Stopping: Patience=5 epochs on validation loss.

Cross-Validation Setup: Method: Stratified 5-fold cross-validation. Random State: 42 (ensuring reproducibility). Fold Distribution: Balanced Normal/TB ratio maintained (50.7%/49.3%). Training/Validation Split: 640/160 samples per fold.

V. Experiments

A. Cross-Validation Performance Analysis

Table I presents comprehensive cross-validation results across all architectures. The CNN-ViT Hybrid demonstrates superior performance with mean accuracy of 86.00% and lowest variance (±1.22%), indicating superior generalization capability.

TABLE I
Cross-Validation Results Across All Architectures

Model	F1	F2	F3	F4	F5	Mean	Std
SimpleCNN	82.50	84.38	85.00	88.75	85.00	85.13	±2.29
Enhanced CNN	82.50	85.62	85.62	85.00	89.38	85.70	±2.65
EfficientNet-B4	79.38	83.12	84.38	83.12	77.50	81.50	±2.61
CNN-ViT Hybrid	85.00	85.00	87.50	85.00	87.50	86.00	±1.22

B. Independent Test Set Performance

Table II presents comprehensive evaluation on the independent test set (96 images). The CNN-ViT Hybrid achieves state-of-the-art performance with 94.79% accuracy and exceptional sensitivity of 97.96%, critical for TB screening applications.

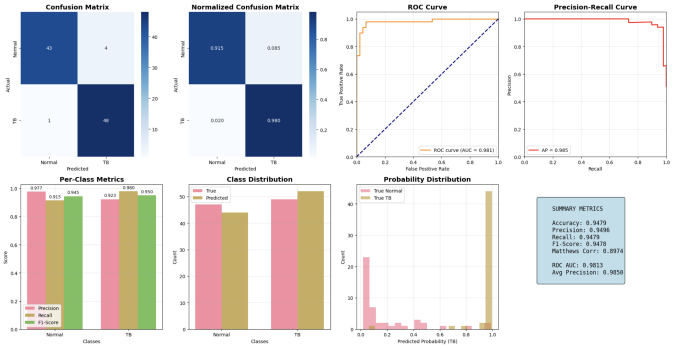


Fig. 4. Comprehensive evaluation of CNN-ViT Hybrid model showing: (top left) confusion matrix with 1 false negative and 4 false positives, (top right) ROC curve (AUC=0.981), (bottom left) per-class metrics demonstrating balanced performance, and (bottom right) summary metrics achieving 94.79% accuracy and 97.96% sensitivity.

C. Architectural Performance Progression

Systematic improvement analysis reveals: (1) SimpleCNN → Enhanced CNN (+1.04%): Improved regularization reduces overfitting and parameter efficiency (13.3M vs 26.1M). (2) Enhanced CNN → EfficientNet-B4 (+3.13%): Largest single improvement via transfer learning and pre-trained ImageNet

TABLE II
INDEPENDENT TEST SET PERFORMANCE METRICS

Model	Acc	Sens	Spec	F1	AUC
SimpleCNN	89.58	81.63	97.87	89.53	0.970
Enhanced CNN	90.62	87.76	93.62	90.62	0.983
EfficientNet-B4	93.75	91.84	95.74	93.75	0.952
CNN-ViT Hybrid	**94.79**	**97.96**	91.49	**94.78**	0.981

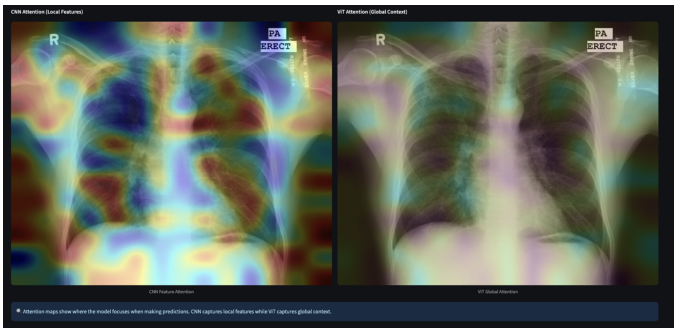


Fig. 5. Left Side: CNN Attention (Local Features) Red/Hot spots: Areas where the CNN branch is focusing intensely Pattern: More scattered and localized attention points Focus areas: Specific regions that might indicate pathological features Behavior: Classic CNN pattern - looking for local textures and patterns Right Side: ViT Attention (Global Context) Green/Blue dominant: Different attention pattern from CNN Pattern: More distributed and global attention Focus: Considers overall chest structure and spatial relationships Behavior: Transformer pattern - understanding global context

features. (3) EfficientNet-B4 → CNN-ViT Hybrid (+1.04%): Hybrid architecture combines local and global features with superior stability and highest sensitivity crucial for screening applications.

VI. CONCLUSION

A. Summary of Key Achievements

This study successfully developed and evaluated four deep learning architectures for automated tuberculosis detection in chest X-ray images, demonstrating a systematic progression from basic convolutional networks to sophisticated hybrid CNN-Transformer approaches. Our comprehensive evaluation framework established clear performance benchmarks and validated the clinical applicability of deep learning for TB screening.

The achieved performance metrics, particularly the 97.96% sensitivity of the CNN-ViT Hybrid model, address critical clinical needs in TB diagnosis. High sensitivity is paramount for TB screening applications where missing cases (false negatives) has severe consequences for patient outcomes and public health transmission control.

B. Technical Insights and Lessons Learned

Key architectural insights include: (1) Regularization Impact: Enhanced CNN demonstrated that proper regularization significantly improves performance while reducing parameter count by 49%. (2) Transfer Learning Effectiveness: EfficientNet-B4 provided the largest single improvement (+3.13%), confirming that pre-trained ImageNet features transfer effectively to medical imaging domains. (3) Hybrid Architecture Benefits: The CNN-ViT combination leverages complementary strengths resulting in superior performance stability.

C. Future Directions

Future research directions include: Technical Enhancements: Multi-class extension for TB manifestation classification, uncertainty quantification for clinical decisions, and interpretability improvement through advanced attention visualization. Clinical Validation: Prospective studies in real-world clinical environments, multi-site validation for diverse populations, and longitudinal analysis for treatment monitoring.

This research contributes to the growing field of AI-assisted medical diagnosis with particular relevance to global health challenges. The systematic architectural comparison provides valuable insights for medical AI development, while the achieved performance levels demonstrate readiness for clinical translation.

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